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1. Context of testing

1.1 Project/test levels/test types:

Project Identification: This test plan identifies the testing strategies for two distinct target applications (Systems Under Test):

- **SUT 1 (OrangeHRM):** A web-based HRM system used for Black-Box and Automation testing.
- **SUT 2 (Library-DB-Express):** A backend Node.js application used for White-Box testing.

Test Levels:

- **Unit Testing:** This level will be performed on SUT 2. The goal is to verify individual backend functions and API endpoints in isolation to ensure code-level correctness.
- **System Testing:** This level will be performed on SUT 1. The application will be tested as a complete, integrated system to validate end-to-end business requirements through the user interface.

Test Types:

- **Black-Box Testing (Functional):** Focuses on testing the requirements of OrangeHRM without internal code knowledge. The following techniques will be applied:

- **Equivalence Class Partitioning (ECP) & Boundary Value Analysis (BVA):** For validating input fields and data constraints in the PIM, and Leave modules.

- **Use Case Testing:** To verify the "Add Employee" workflow within the PIM module.

- **Decision Tables:** To test the business logic of the "Leave Request/Approval" process in the Leave module.

- **Performance, Load, Stress, and Endurance (Soak) Testing:** Performed on SUT 1 to evaluate responsiveness, stability under peak loads, and long-term resilience under continuous traffic. This ensures the application can handle expected traffic and sustained sessions without performance degradation or connection exhaustion
- **Compatibility Testing:** Performed on **SUT 1** to ensure the application's core functionalities (Login, PIM) remain consistent across different web browsers (e.g., Chrome, Firefox, Edge).

- **Usability Testing:** A heuristic-based evaluation of **SUT 1** to assess user-friendliness, navigation efficiency, and the clarity of system feedback during key workflows.
- **Automated Testing:** Selenium will be used to automate repetitive functional test cases. Most tests will be designed using Selenium IDE, while at least two key scenarios (e.g., Login and Add Employee) will be exported and executed as scripts using Selenium WebDriver to demonstrate scripting proficiency.
- **White-Box Testing (Structural):** Focuses on the internal logic and structure of the SUT 2 source code. It includes:
 - **Unit Testing:** API endpoints and backend functions will be tested using Jest and Supertest to ensure HTTP request/response cycles are handled correctly.
 - **Security Testing:** Ensuring basic authentication mechanisms and authorized access to data by verifying API endpoint protection, input validation, error handling, and resource access control through URL parameters.
 - **Coverage & Mutation Analysis:** Measuring test thoroughness through statement coverage and using the Stryker tool for mutation testing to evaluate test suite robustness.

1.2 Test items:

Item 1: OrangeHRM Open Source Application

- **Version/Revision:** OrangeHRM OS 5.8
- **Reference:** <https://opensource-demo.orangehrmlive.com/>
- **Purpose:** This web-based application is used as the target for system-level functional testing. The focus will be on the Login mechanism, PIM (Personnel Information Management), and Leave modules to verify business workflows and GUI consistency.

Item 2: Library-DB-Express Source Code

- **Version/Revision:** v1.0 (Course Project Repository)
- **Reference:** Source code provided by the lecturer for the SE 2226 term project.
- **Purpose:** This backend API and database system is the target for structural white-box testing. The focus is to ensure code quality through script-based unit tests using Jest/Mocha, security checks, and mutation analysis using Node.js/Express.

1.3 Test scope:

Features to be Tested (In-Scope):

A. OrangeHRM (SUT 1):

- **Functional Modules:**

- **User Authentication:** Secure login and logout processes.
- **PIM Module:** Verification of the full employee lifecycle, focusing on "Add Employee" for record creation and the "Employee List" for searching and managing personal details.
- **Leave Module:** End-to-end management of the leave process, including "Apply" for user requests, "Entitlements" for checking leave balances, and the "Approve/Reject" supervisor.

- **Design-Based Testing:**

- **Input Validation:** Applying **Equivalence Class Partitioning (ECP)** and **Boundary Value Analysis (BVA)** to critical fields like Employee ID, Names, and Dates.
- **Logical Consistency:** Verification of complex leave workflows and PIM workflows using **Decision Tables** and **Use Case Analysis**.

- **Non-Functional & Automation:**

- **GUI Automation:** Scripting of critical paths (Login, Add Employee) using **Selenium IDE** and **WebDriver**.
- **Concurrency & Performance:** Load, Stress, and Endurance (Soak) testing to evaluate system stability, response times, and long-term resilience under high and sustained user traffic.
- **Cross-Browser Compatibility:** Ensuring consistent behavior on **Chrome, Firefox, and Edge**.
- **Usability & UX Validation:** Heuristic evaluation to ensure intuitive navigation and helpful system feedback/error messages.

B. Library-DB-Express (SUT 2) - White-Box & Structural Focus

- **API Verification:** Validation of backend communication for all CRUD operations (GET, POST, PUT, DELETE).

- **Structural Integrity:**

- **White-Box Logic:** Detailed verification of internal code paths and decision branches.
- **Code Coverage:** Achieving target percentages for **Statement and Branch coverage** using Jest/Istanbul.

- **Quality & Robustness:**

- **Mutation Testing:** Evaluating test suite effectiveness and detecting surviving mutants using **Stryker**.
- **Security Logic:** Ensuring API endpoint protection, input validation, error handling, and resource access control through URL parameters.

Features NOT to be Tested (Out-of-Scope):

- **Full-Scale Penetration Testing:** While **security audits and vulnerability assessments** (such as XSS and input validation) are performed on SUT 2, exhaustive network-level penetration testing or external cyber-attack simulations are excluded.
- **Visual/Aesthetic Testing:** Purely cosmetic UI elements (specific color hex codes, exact font sizes) are excluded, while functional visibility and layout consistency remain in-scope for Usability and Compatibility analysis.

1.4. Test basis:

The following resources, academic materials, and technical documentations serve as the primary basis for the design, implementation, and execution of the tests within this project:

- **Project Guidelines:** The official "SE 2226 Term Project" instruction document provided by Doç. Dr. Korhan Karabulut, which defines the requirements for both functional and structural analysis.
- **SE 2226 Course Materials:** Academic lecture notes and slides provided during the semester, specifically focusing on:

-**Black-Box Methodologies:** Equivalence Class Partitioning (ECP), Boundary Value Analysis (BVA), Use Case Testing, and Decision Table Methods.

-**White-Box Methodologies:** Unit Testing principles, Statement and Branch Coverage requirements, and Mutation Analysis logic.

- **System Functional Specifications:**

-The live functional behavior and user interface of the OrangeHRM Open Source Demo (v5.8), which serves as the requirement source for Black-Box test cases.

-The backend API architecture and source code of the Library-DB-Express application, which serves as the basis for White-Box unit tests.

- **International Standards:** ISO/IEC/IEEE 29119 (Software Testing Standard) and IEEE 829 (Standard for Software Test Documentation) are used as the structural framework for this test plan.
- **Tool Documentation:** Official technical guides for Selenium IDE/WebDriver (for GUI automation scripts) and Stryker (for mutation testing analysis).

2. Assumptions and constraints

Assumptions:

- **System Availability:** It is assumed that both the OrangeHRM public demo and the Library-DB-Express source code will be consistently accessible throughout the project duration.
- **Source Code Integrity:** It is assumed that the provided backend source code is functionally complete and allows for the execution of unit tests without requiring external proprietary licenses.
- **Technical Capability:** It is assumed that the project team possesses the necessary technical proficiency to operate testing tools such as Selenium WebDriver, Stryker, Jest, and Supertest.
- **Environmental Stability:** It is assumed that the local development environments (Node.js, Java, Chrome Driver) are correctly configured to support the execution of the test scripts.

Constraints:

- **Project Deadline:** The most significant constraint is the final submission and presentation date, fixed for May 8th, 2026.
- **Budgetary Limits:** As an academic exercise, the project is restricted to open-source and free-tier tools. No budget is allocated for commercial testing platforms.
- **Scope Boundaries:** Testing is focused on functional modules, structural logic, and specific non-functional attributes (Performance, Usability, Compatibility). Additionally, the scope includes security auditing and vulnerability analysis to evaluate the software's resilience against common web threats and ensure high-quality security metrics.
- **Network Dependencies:** Testing the OrangeHRM system is dependent on a stable internet connection, as it is a cloud-hosted demo environment.
- **Regulatory Standards:** All testing activities must adhere to the academic integrity rules defined by the university and the specific guidelines provided for the SE 2226 course.

3. Test strategy

3.1 General:

The testing strategy is designed to validate both functional requirements and structural integrity. A Risk-Based Testing approach is adopted for SUT 1 (OrangeHRM), focusing on high-priority business workflows such as user management and leave processing. For SUT 2 (Library-DB-Express), a Structural Testing approach is used to ensure code robustness through automated unit testing and mutation analysis.

3.2 Test levels:

- **Unit Testing:** Performed on SUT 2 to verify individual backend functions, API route controllers, and database logic in isolation.
- **System Testing:** Performed on SUT 1 to validate the end-to-end functionality of the integrated application through the user interface (GUI).

3.3 Test types:

- **Black-Box Testing (Functional):** Focused on validating SUT 1 requirements without looking at the internal code, using techniques like ECP, BVA, and Decision Tables.
- **White-Box Testing (Structural):** Focused on the internal paths and logic of SUT 2, including branch coverage and mutation testing.
- **Automated Testing:** Utilizing Selenium for GUI regression and Jest/Supertest for backend unit verification.
- **Non-Functional Testing:** Includes Performance, Load, Stress, and Endurance (Soak) testing for SUT 1 to ensure the system remains stable, responsive, and resilient during both peak traffic and extended periods of concurrent user activity.
- **Design & Compatibility Testing:** Includes Usability and Compatibility testing for SUT 1 to verify cross-browser stability and user-friendly navigation.
- **Security Testing (SUT 2):** Focused on identifying vulnerabilities within the backend logic, such as information disclosure via stack traces, resource access control, and endpoint resilience against malformed inputs.

3.4 Test deliverables:

The following test documentation and technical artifacts will be delivered throughout the project lifecycle, in accordance with the project requirements and ISO/IEC/IEEE 29119 standards:

- **Master Test Plan:** The strategic guiding document (this document) defining the scope and approach.
- **Test Model Specification:** Documentation of the logical models used for testing, including:
 - **Black-Box Models (SUT 1):** Equivalence Partitioning (EP), Boundary Value Analysis (BVA), Use Case Testing, and Decision Tables for OrangeHRM modules.
 - **White-Box Models (SUT 2):** Logic analysis focused on ensuring Statement and Branch Coverage for the Library-DB-Express backend.
- **Test Case Specification:** Detailed manual test cases and automated test steps derived from the test models.

- **Performance & Endurance Analysis Reports:** Quantitative reports for SUT 1 detailing response times, throughput, and system behavior under stress, including long-term stability analysis (Soak Testing) to identify resource exhaustion.
- **Usability Evaluation Report:** A document detailing the findings of heuristic evaluations and UX improvements for SUT 1.
- **Compatibility Matrix:** A cross-browser testing report showing functional consistency across Chrome, Firefox, and Edge.
- **Security Audit & Vulnerability Report:** A technical document detailing SUT 2 backend resilience, focusing on input validation, error handling (stack trace leaks), and API endpoint security.
- **Automated Test Scripts:** The source code and execution files:
 - **SUT 1:** Selenium IDE (.side) files and exported WebDriver scripts.
 - **SUT 2:** JavaScript-based unit test scripts and API validation code.
- **Integrated Incident Reports:** Formal records of discovered defects and anomalies, documented within their respective technical reports (e.g., Performance, Security, and Mutation reports), including severity levels and impact analysis.
- **Coverage & Mutation Reports:** Quantitative analysis reports showing:
 - Statement and Branch coverage percentages.
 - Stryker Mutation Analysis results to verify test suite robustness.
- **Test Completion Report (Final Test Summary Report):** A comprehensive final report summarizing the testing activities, pass/fail ratios, and final quality assessment of both SUT 1 and SUT 2.

3.5 Test design techniques:

- **For SUT 1:**
 - **Functional:** Equivalence Partitioning (EP), Boundary Value Analysis (BVA), Use Case Testing, and Decision Tables.
 - **Non-Functional & UX:** Heuristic Evaluation (for Usability), Cross-Browser Verification (for Compatibility), and Quantitative Performance Modeling (for Load, Stress, and Endurance testing).
- **For SUT 2:** Statement and Branch Coverage (via Jest/Istanbul), Mutation Testing (via Stryker), and Logic-based Security Verification. Security testing specifically focuses on:
 - **Input Validation:** Ensuring resilience against XSS and SQL Injection.
 - **Error Handling:** Verifying that the system prevents information disclosure (e.g., stack trace leaks).

- **Resource Access Control:** Testing endpoint protection against unauthorized URL parameter manipulation.

3.6 Entry and exit criteria:

Entry Criteria: System test execution shall commence only when the following conditions are met:

- **Approval:** The Master Test Plan is finalized and signed off by the project team.
- **Documentation:** All test cases (Black-Box and White-Box) are documented and approved.
- **Environment Readiness:** The SUT 1 (OrangeHRM) demo environment is accessible, and the SUT 2 (Library-DB-Express) local environment is correctly configured with Node.js and SQLite.
- **Test Data:** Synthetic datasets for ECP, BVA, and security payloads (XSS/SQLi) are prepared and ready for injection.

Exit Criteria: Testing activities are considered complete when:

- **Functional Completion:** 100% of high-priority test cases for SUT 1 have been executed, with all "Critical" and "High" severity defects resolved.
- **Structural Integrity (SUT 2):** A minimum of **95% Statement Coverage** and **90% Branch Coverage** is achieved (Reflecting the high quality verified in the White-Box Report).
- **Robustness Verification:** The mutation score must meet the defined threshold, confirming that the test suite is capable of detecting logic faults.
- **Security & Risk Assessment:** All identified security vulnerabilities (specifically Stack Trace leaks and DoS risks) are documented with a final impact analysis.
- **Defect Management:** Any remaining Severity 2 or 3 defects are reviewed and logged in the Final Test Summary Report for future maintenance.

3.7 Test completion criteria:

Testing is finalized when the Exit Criteria are met and the Final Test Summary Report is signed off for presentation.

3.8 Metrics to be collected:

- **Test Case Execution Rate:** The percentage of planned test cases that have been executed.
- **Test Case Pass Rate:** The ratio of passed tests to total tests executed, tracking the overall functional reliability.

- **Defect Severity Distribution:** The classification of identified issues by impact (Critical, High, Medium, Low) to prioritize risk management.
- **Code Coverage (SUT 2):** Quantitative measurement of Statement and Branch coverage via **Jest/Istanbul** to ensure structural thoroughness.
- **Mutation Score (SUT 2):** Evaluation of the test suite's robustness and fault-detection capability using **Stryker**.
- **Performance Benchmarks (SUT 1):** Measurement of Average Response Time, Throughput, and Error Rates to evaluate system stability under different load conditions.
- **System Breaking Point:** Identification of the load level where system performance degrades beyond acceptable reliability thresholds.

3.9 Test data requirements:

This section specifies the origin, characteristics, and management of the data required for both SUTs:

- **Origin of Data:**
 - * **SUT 1:** Test data is derived from the **Equivalence Class Partitioning (ECP)** and **Boundary Value Analysis (BVA)** tables.
 - **SUT 2:** Data is sourced from the backend project repository provided by the lecturer.
- **Specific Data Requirements (SUT 1):**
 - **Employee ID:** Numeric values between 1 and 10 digits (e.g., "5421").
 - **Full Names:** Alphabetic strings with a length of 2 to 50 characters, including Unicode support for non-Latin characters.
 - **Photos:** Image files meeting the criteria of .jpg, .png, or .gif formats with a size limit of 1MB.
- **Business Logic Data:** Conditions for leave requests based on processed status, leave type, date validity, and balance sufficiency as defined in the **Decision Table**.
- **Data Confidentiality:** Since open-source and demo environments are used, no real-world sensitive personal data is required; all data is synthetic/mock.
- **Responsibility:** The Test Analyst and SDET Specialist are responsible for preparing and maintaining the test datasets.

3.10 Test environment requirements:

To ensure the consistent execution of tests and the reliability of results, the following environment requirements must be met:

- **Hardware:** Standard workstation or laptop (Minimum 8GB RAM recommended for running automation suites and mutation analysis simultaneously).
- **Operating System:** Cross-platform compatibility (Windows 10/11, macOS, or Linux).
- **Browsers:** Multi-browser environment including **Google Chrome, Mozilla Firefox, and Microsoft Edge** (Latest versions) to ensure cross-browser compatibility. Corresponding ChromeDriver are required for Selenium WebDriver execution.
- **Runtime Environments:**
 - * **Node.js (LTS Version):** Required for SUT 2 and the Jest/Supertest frameworks.
 - **NPM (Node Package Manager):** For managing project dependencies and test tools.
- **Database:** SQLite (as integrated within the Library-DB-Express backend) for data persistence testing.
- **Testing Tools:**
 - * **Selenium IDE:** For initial GUI test recording (SUT 1).
 - **Selenium WebDriver:** For scripted automated testing (SUT 1).
 - **Jest:** For unit and integration testing (SUT 2).
 - **Stryker:** For mutation testing and robustness analysis (SUT 2).
 - **Supertest:** Integration testing for REST APIs and simulating HTTP requests to verify endpoint responses.
- **Development Tools:** Visual Studio Code for developing backend unit tests, debugging Node.js scripts, and managing the SUT 2 test environment.
- **Performance Tools:** JMeter is utilized for executing Load, Stress, and Endurance (Soak) tests on SUT 1. These tools measure system stability and resource consumption during both peak traffic and extended operational periods.
- **Connectivity:** High-speed stable internet access is mandatory for accessing the OrangeHRM Cloud Demo environment.

3.11 Testing activities and estimates:

The testing implementation is described through a Work Breakdown Structure (WBS) with associated effort estimates:

Activity (WBS)	Description	Estimated Effort
Test Planning	Finalizing the Master Test Plan and strategy.	10 Hours
Test Analysis & Design	Designing test cases using ECP, BVA, and Use Case models.	23 Hours
Scripting (SUT 1)	Developing Selenium IDE and WebDriver scripts.	20 Hours
Non-Functional Testing	Conducting Performance, Load, Stress, Soak, Usability, and Compatibility tests.	18 Hours
Backend Testing (SUT 2)	Writing Jest/Supertest unit tests and security validation code.	15 Hours
Structural Analysis & Security Verification	Executing Stryker for mutation testing, measuring coverage, and performing security logic verification.	8 Hours
Environment Setup	Configuring Node.js, Selenium, and SQLite environments.	5 Hours
Reporting & Demo	Creating the Test Completion Report and project slides.	10 Hours
Total Estimated Effort		109 Hours

Budget: No financial budget is allocated as all tools used (Selenium, Jest, Stryker) are open-source.

4. Staffing

4.1 General:

The testing team consists of four specialized members collaborating to ensure the quality of both SUT 1 (OrangeHRM) and SUT 2 (Library-DB-Express). While specific roles such as **Automation Engineer** and **White-Box SDET** are assigned to lead each activity, a cross-functional collaboration model is adopted. This ensures that all team members support each other during test analysis, script development, and documentation phases to maintain high quality and meet project deadlines.

4.2 Roles and responsibilities:

The roles and corresponding responsibilities for the testing activities are distributed as follows:

- **Test Lead & Management**

- **Primary Leader:** Helin Güler
- **Secondary Leader:** Aslı Zeynep Akhan
- **Responsibilities:** Responsible for overall project coordination, assigning tasks, risk management, and the final approval of all test artifacts.

- **Documentation Specialist**

- **Lead:** Aslı Zeynep Akhan
- **Responsibilities:** Leading the creation and maintenance of the Master Test Plan, ensuring the document is updated iteratively, and consolidating final reports.

- **Test Analyst (Black-Box & Design)**

- **Lead:** Helin Güler & Şevval Demir
- **Responsibilities:** Analyzing SUT 1 requirements. Helin focuses on **ECP** and **BVA** techniques, while Şevval leads the development of **Decision Tables** and **Use Case Testing**.

1. SUT 1 (OrangeHRM) Focus Areas

- **Automation Engineer**

- **Lead:** Deniz Berfin Acar
- **Responsibilities:** Responsible for creating automated GUI scripts using Selenium IDE and exporting key scenarios to WebDriver scripts. This involves transforming ECP, BVA, and Use Case tables into executable scripts.

- **Performance & Load Analyst**

- **Lead:** Helin Güler
- **Responsibilities:** Responsible for conducting performance, load, stress, and endurance (soak) testing to identify system bottlenecks, long-term stability issues, and breaking points under sustained stress.

- **Compatibility Specialist**

- **Lead:** Aslı Zeynep Akhan
- **Responsibilities:** Responsible for verifying system stability and visual consistency across different browsers (Chrome, Firefox, Edge) to ensure cross-browser compatibility.

- **Usability Specialist**

- **Lead:** Şevval Demir

- **Responsibilities:** Responsible for performing heuristic evaluations and analyzing user-friendliness, navigation flow, and the effectiveness of system error messages.

2. White-Box SDET Specialist (SUT 2 - Library-DB-Express)

Responsibilities: Overall owner of the structural testing process for SUT 2. This role is subdivided into the following technical focus areas:

- **LDB Unit & White-Box Logic**

- **Lead:** Şevval Demir
- **Focus:** Managing the Node.js/NPM environment setup and developing backend unit tests to verify internal code paths and CRUD operations using Jest and Supertest.

- **LDB Code Coverage Analysis**

- **Lead:** Helin Güler
- **Focus:** Measuring and reporting the statement and branch coverage percentages. The goal is to ensure the test suite meets the project's coverage requirements.

- **Mutation & Security Testing**

- **Lead:** Aslı Zeynep Akhan
- **Focus:** Conducting security audits to verify input validation and error handling, while performing mutation testing to ensure the robustness and fault-detection capability of the test suite.

3. Documentation & Workflow Notes

- **Workflow Note:** The results from **Code Coverage** and **Mutation Testing** must be reported back to the Unit Test developer (**Şevval Demir**). This feedback loop is essential for identifying weak points and refactoring for higher quality.
- **Incident Reporting:** Each team member is responsible for documenting bugs found within their specific focus area.
- **Final Integration:** **Deniz Berfin Acar** is responsible for compiling all individual incident reports and technical results into the final **Test Completion Report** and **Presentation Slides**.

Test Completion Report & Analysis

- **Lead:** Deniz Berfin Acar
- **Responsibilities:** Consolidating all test results, bug reports, and coverage metrics into the final completion report; ensuring all exit criteria are documented and met.

Final Presentation & Slides

- **Lead:** Deniz Berfin Acar
- **Responsibilities:** Designing the project presentation slides, coordinating the demo walkthroughs for SUT 1 and SUT 2, and managing the final delivery to the instructor.

4.3 Schedule:

The following schedule summarizes the overall testing activities and milestones. The timeline is designed to allow for iterative updates to the test plan as analysis and coding progress.

Activity / Milestone	Start Date	End Date	Deliverables / Outcomes
Phase 1: Test Planning	April 6, 2026	April 19, 2026	Finalized Master Test Plan & Strategy
Phase 2: Test Analysis & Design	April 11, 2026	April 19, 2026	Test Case Specifications & Design Tables
Phase 3: Environment Setup	April 18, 2026	April 21, 2026	Ready Selenium & Node.js/SQLite environments
Phase 4: Structural & Coverage Testing (SUT 2)	April 20, 2026	April 22, 2026	Statement & Branch Coverage Report , Unit Test Execution
Phase 5: Advanced Technical Testing (SUT 1 & 2)	April 21, 2026	April 30, 2026	Security Audit Report, Mutation Analysis Report , Automated Script Execution
Phase 6: Non-Functional & UX Analysis	May 1, 2026	May 4, 2026	Performance & Endurance Report, Usability & Compatibility Report
Phase 7: Final Reporting & Submission	May 5, 2026	May 8, 2026	Test Completion Report (Summary) & Final Presentation Slides